

# Final Project: Stock Market Data Pipeline & Analysis

## 1. Project Overview

In this project, students will design and implement an end-to-end data engineering pipeline that ingests, processes, and analyzes stock market data.

The pipeline will collect historical and incremental stock data using Python, clean and transform the data, store it in a structured database, and generate analytical insights to support financial decision-making.

This project simulates a real-world scenario where financial teams rely on reliable and well-structured data to monitor market trends and evaluate investment opportunities.

## 2. Business Objective

The goal of this project is to enable data-driven analysis of stock performance by:

- Tracking price trends over time
- Comparing performance across multiple companies
- Measuring volatility and risk
- Supporting basic investment insights

## 3. Data Source

- Source: Yahoo Finance
- Access Method: Python (`yfinance` library)
- Data Frequency: Daily stock data

### Selected Companies (Example)

Students should work with at least 8–10 companies, such as:

- AAPL, MSFT, AMZN, GOOGL, META, TSLA, NVDA

- **Data Columns**
- Date
- Open
- High
- Low
- Close
- Volume

## 4. Project Requirements

### 4.1 Data Ingestion

Students must:

- Use Python to fetch stock data using `yfinance`
- Retrieve historical data for selected companies
- Implement logic to support incremental updates (new data only)

### 4.2 Data Cleaning

Students are required to:

- Standardize the date format (YYYY-MM-DD)
- Remove duplicate records
- Validate numerical columns (prices and volume)
- Handle missing values appropriately

Note: Missing dates due to weekends or holidays should be understood and handled correctly (not treated as errors).

### 4.3 Data Transformation

Students must enrich the dataset by calculating:

- **Daily Return**  
Percentage change in closing price compared to the previous day
- **Price Range**  
Difference between high and low prices
- **Moving Average (e.g., 7-day)**  
Rolling average of closing price

- **Volatility (7-day)**  
Standard deviation of daily returns

## 4.4 Data Modeling (PostgreSQL)

Students must design and implement the following schema:

### **fact\_stock\_prices**

- date
- ticker
- open
- high
- low
- close
- volume
- daily\_return

Primary Key: (date, ticker)

### **dim\_company**

- ticker
- company\_name

Requirements:

- Proper primary keys
- No duplicate records
- Clear and consistent schema design

## 4.5 Incremental Loading

Students must implement:

- Logic to load only new records (based on date)
- Prevention of duplicate inserts
- Idempotent pipeline behavior (safe re-runs)

## 4.6 Orchestration (Airflow)

Students must create an Airflow DAG that includes:

- extract\_stock\_data

- transform\_data
- load\_to\_database

Requirements:

- Task dependencies
- Basic retry mechanism
- Manual or scheduled execution

## **4.7 Data Validation**

Students must ensure:

- No duplicate records for the same date and ticker
- No null values in key fields
- Consistency of calculated columns

## **4.8 Analytical Queries**

Students must write SQL queries to answer:

- Which stock performed best over a given period?
- Which stock is the most volatile?
- What is the average closing price per company?
- What are the price trends over the last 30 days?

## **5. Deliverables**

Students must submit:

1. Python ETL pipeline (modular and organized)
2. PostgreSQL database with final tables
3. SQL scripts (table creation and analytical queries)
4. Airflow DAG file
5. README document
6. Github repo with all the mentioned above.

## 6. README Requirements

The README must include:

- Description of the pipeline architecture
- Data cleaning decisions and assumptions
- Incremental loading logic
- Explanation of transformations
- Any limitations or challenges faced